Chapter Review

Key Terms

commission A commission is a percentage of total sales as determined by the rate of commission.
discount An amount of discount is a percent off the original price, determined by the discount rate.
mark-up The mark-up is the amount added to the wholesale price, determined by the mark-up rate.
percent A perfecnt is a ratio whose denominator is 100.
percent decrease The percent decrease is the percent the amount of decrease is of the original amount.
percent increase The percent increase is the percent the amount of increase is of the original amount.
proportion A proportion is an equation of the form $\frac{a}{b} = \frac{c}{d}$, where $b \neq 0$, $d \neq 0$.The proportion states two ratios or rates are equal. The proportion is read "a is to b, as c is to d".
sales tax The sales tax is a percent of the purchase price.
simple interest If an amount of money, P, the principal, is invested for a period of t years at an annual interest rate r, the amount of interest, I, earned is $I = Prt$. Interest earned according to this formula is called simple interest.

Key Concepts

6.1 Understand Percent

- **Convert a percent to a fraction.**
  Step 1. Write the percent as a ratio with the denominator 100.
  Step 2. Simplify the fraction if possible.
- **Convert a percent to a decimal.**
  Step 1. Write the percent as a ratio with the denominator 100.
  Step 2. Convert the fraction to a decimal by dividing the numerator by the denominator.
- **Convert a decimal to a percent.**
  Step 1. Write the decimal as a fraction.
  Step 2. If the denominator of the fraction is not 100, rewrite it as an equivalent fraction with denominator 100.
  Step 3. Write this ratio as a percent.
- **Convert a fraction to a percent.**
  Step 1. Convert the fraction to a decimal.
  Step 2. Convert the decimal to a percent.

6.2 Solve General Applications of Percent

- **Solve an application.**
  Step 1. Identify what you are asked to find and choose a variable to represent it.
  Step 2. Write a sentence that gives the information to find it.
  Step 3. Translate the sentence into an equation.
  Step 4. Solve the equation using good algebra techniques.
  Step 5. Write a complete sentence that answers the question.
  Step 6. Check the answer in the problem and make sure it makes sense.
- **Find percent increase.**
  Step 1. Find the amount of increase:
  $\text{increase} = \text{new amount} - \text{original amount}$
  Step 2. Find the percent increase as a percent of the original amount.
- **Find percent decrease.**
  Step 1. Find the amount of decrease.
  $\text{decrease} = \text{original amount} - \text{new amount}$
  Step 2. Find the percent decrease as a percent of the original amount.

6.3 Solve Sales Tax, Commission, and Discount Applications

- **Sales Tax** The sales tax is a percent of the purchase price.
  - $\text{sales tax} = \text{tax rate} \cdot \text{purchase price}$
  - $\text{total cost} = \text{purchase price} + \text{sales tax}$
- **Commission** A commission is a percentage of total sales as determined by the rate of commission.
  - $\text{commission} = \text{rate of commission} \cdot \text{original price}$
- **Discount** An amount of discount is a percent off the original price, determined by the discount rate.